

Customer Agent OS Lite

V1 Product Contract: Personas, Journeys, Outcomes, Approvals, Screens, and Acceptance Criteria

Product definition document - business and user experience scope only

Purpose. Provide a stable, readable product foundation before technical architecture, framework selection, or implementation planning.

Project. Customer Agent OS Lite - a lightweight platform for building and operating domain-specific customer-facing AI agents, proven first through an e-commerce support agent.

Status. Product planning baseline. Changes should be intentional and documented rather than introduced casually during engineering.

Prepared for: Product analysis, portfolio planning, and principal-level engineering design

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1. Purpose, scope, and contract rules

Why this document exists

The fixed feature document defines the complete product blueprint. This product contract now makes the first customer journeys, personas, outcomes, approval boundaries, screens, and acceptance criteria precise enough to become the input for technical design.

1.1 V1 contract scope

- Primary scope: the e-commerce customer support agent and the human support workspace.
- Supporting scope: the minimum operations screens required to configure policies, knowledge, journeys, and users for those experiences.
- Channels: web chat only for V1.
- Domain: e-commerce orders, returns, replacements, refunds, subscriptions, and escalation.
- Business systems: simulated or sandbox commerce systems are acceptable for the portfolio build, but product behavior must match real business operations.

1.2 Contract rules

- The customer interacts with one primary support agent, even when internal specialist capabilities are used.
- The agent may explain or propose an action without approval, but consequential account or financial changes require confirmation.
- The agent never reports success unless the relevant business system confirms success.
- A failed or uncertain case moves to a clear recovery or human handoff path.
- The customer should not repeat information already supplied unless identity, security, or correctness requires reconfirmation.
- Threshold values are tenant-configurable; V1 demo defaults are defined in this document to remove ambiguity.

V1 demo policy defaults

Automated refund limit: \$100. Automated low-risk replacement limit: \$250. Damaged-item evidence required for items above \$75. These defaults demonstrate the product behavior and can later be changed per tenant.

2. V1 personas

2.1 End customer

Person seeking help with an order, return, replacement, refund, or subscription.

Resolve an issue quickly, understand the decision, retain control over consequential actions, and reach a human without repeating the story.

Dimension	Fixed product contract
Core jobs	Ask a question; identify an account or order; provide missing information; upload evidence; choose an option; confirm an action; view progress; request a person.
Allowed actions	View only their own account data; confirm or cancel proposed

Dimension	Fixed product contract
	actions; choose among permitted options; request escalation at any point.
Constraints	Must complete identity verification for protected data or actions; cannot bypass policy or approval rules; cannot access internal notes.
Success condition	The issue is completed or cleanly handed to a human with transparent status and no unnecessary repetition.

2.2 Human support representative

Person who receives escalations, handles exceptions, approves permitted actions, and completes cases.

Understand the case immediately, act with complete context, avoid duplicated work, and correct the agent when needed.

Dimension	Fixed product contract
Core jobs	Review queue; inspect case; join conversation; approve or reject; edit a response; perform manual action; return control; resolve; classify failure.
Allowed actions	Access assigned customer context and internal support information; perform actions allowed by role; approve only within assigned authority.
Constraints	Must follow tenant permissions and policy; cannot silently alter immutable audit history; must record a resolution reason.
Success condition	The case is resolved efficiently, the customer remains informed, and useful correction data is captured.

2.3 Support operations manager

Business owner responsible for support journeys, policies, quality, and operating outcomes.

Configure safe customer experiences, update policies and knowledge, review performance, and improve containment without harming trust.

Dimension	Fixed product contract
Core jobs	Manage supported journeys; edit thresholds; connect knowledge; inspect gaps; define queues; review evaluations and releases; inspect trends.
Allowed actions	Configure business rules and content within assigned tenant and environment; cannot grant themselves security permissions outside their role.
Constraints	Changes require versioning and, for production, review or release controls; sensitive security settings belong to administrators.

Dimension	Fixed product contract
Success condition	Support behavior is current, measurable, safe, and aligned with business outcomes.

2.4 Tenant administrator

Organization owner responsible for access, security boundaries, environments, budgets, and governance.
Keep tenant data and actions controlled while allowing product and support teams to operate effectively.

Dimension	Fixed product contract
Core jobs	Manage users, roles, environments, approved models, retention, quotas, budgets, and approval authorities.
Allowed actions	Full tenant administration within organizational ownership; can assign but not bypass system-enforced audit and isolation controls.
Constraints	High-impact administrative changes require explicit confirmation; actions must be auditable.
Success condition	Only authorized users, models, knowledge, and actions are available in each environment.

2.5 Developer or integration engineer

Technical user extending tools, sandbox services, integrations, tests, and agent configurations.
Build reusable capabilities, test them safely, inspect failures, and deploy changes through a controlled release process.

Dimension	Fixed product contract
Core jobs	Register tools; configure sandbox integrations; author tests; inspect traces; use local runner; propose releases.
Allowed actions	Development and staging access by default; production write access only when explicitly granted.
Constraints	Cannot bypass customer confirmation, policy, approval, or audit controls through custom tooling.
Success condition	New capabilities are reusable, testable, observable, and safe for release.

3. Shared journey rules and business outcomes

3.1 Shared journey rules

Rule	Fixed product behavior
Identity	General policy questions may be answered anonymously. Account-specific data or changes require customer identification and step-up verification when risk warrants it.
Clarification	The agent asks only for information needed to progress the selected journey.
Evidence	Policy or eligibility decisions show the governing source or a concise policy explanation.
Preview	Consequential actions display a summary of the target, amount or effect, timing, and next step before confirmation.
Confirmation	The customer must explicitly confirm returns, refunds, replacements, cancellations, and address changes.
Verified completion	Success is shown only after the business service confirms the action and returns a reference or final state.
Failure	The product distinguishes temporary failure, policy denial, missing information, and human-review requirement.
Handoff	The customer can request a human at any time. The product passes the transcript, summary, state, evidence, and attempted actions.
Progress	Longer work shows a clear status and what event or person is awaited.

3.2 Fixed outcome definitions

Journey	Successful business outcome
Order tracking	The correct order and current shipment state are shown, including next expected event or escalation for abnormal cases.
Return eligibility	A policy-grounded eligible or ineligible decision is shown with reason and relevant deadline or condition.
Create a return	A return authorization exists and the customer receives a return ID, method, instructions, deadline, and label or drop-off details.
Damaged-item replacement	Evidence and eligibility are complete; a replacement is reserved or an approval/human review is pending with clear status.
Refund	The payment or refund system confirms the refund request and returns an amount, method, status, and reference.
Subscription cancellation	The subscription system confirms immediate or end-of-term cancellation and states access and billing consequences.
Human escalation	The case is assigned or queued with complete context, reason, priority, and customer-visible next step.

4. Order tracking

Journey purpose

Give the customer the correct status of an order or shipment and route abnormal situations without making unsupported delivery promises.

Contract area	Fixed decision
Preconditions	The customer has an identifiable account or can provide an order identifier plus verification information.
Required inputs	Order selection or order ID; optional question about a specific shipment or item.
Customer confirmation	No confirmation is required to view tracking information. Confirmation is required before any change such as cancellation, replacement, or address update.
Human approval	No approval for information retrieval. Human review is required for unsupported delivery promises, high-risk address changes, or disputed delivery claims beyond the supported path.
Escalation triggers	Customer asks for a human; order or identity cannot be verified; repeated carrier failure; delivered-but-missing risk; unsupported cross-border or special handling; tool unavailable after retry.
Successful outcome	The customer sees the correct current state and a truthful next step.

Main customer flow

1. Identify or verify the customer when account data is required.
2. List recent orders when the request does not identify one.
3. Ask the customer to select the relevant order.
4. Retrieve fulfillment and carrier status.
5. Explain the latest event, current status, and expected next event.
6. Offer a relevant next action when delayed, delivered-but-missing, or exception status is detected.
7. Close the journey or transition to escalation or another supported journey.

Alternative and failure flows

- Multiple shipments: show each shipment separately and identify which items belong to each.
- No tracking data yet: explain that fulfillment has started but carrier tracking is not available; provide the next expected update.
- Carrier exception: explain the known exception and offer escalation when no supported self-service resolution exists.
- Delivered but customer cannot find package: collect confirmation of address and basic checks, then route to a missing-package path or human review.
- Order not found: allow one retry using another identifier, then escalate or provide account recovery guidance.

Acceptance criteria

PASS The displayed order and shipment belong to the authenticated customer.

PASS The answer includes the latest known carrier or fulfillment event and does not invent an exact date when unavailable.

PASS Multiple shipments are not merged into a misleading single status.

PASS A carrier or order-service failure produces a clear recovery or escalation path.

PASS No customer data from another account or tenant is displayed.

5. Return eligibility

Journey purpose

Determine whether an item can be returned under the policy that applied to the purchase and explain the decision clearly.

Contract area	Fixed decision
Preconditions	The customer and order are identified; the item is selected; applicable policy sources are available.
Required inputs	Order, item, return reason, item condition when relevant, and purchase/fulfillment date from the order system.
Customer confirmation	No confirmation is required to receive an eligibility decision. Confirmation is required to proceed with return creation.
Human approval	No human approval for standard eligibility. Human review is required for policy conflicts, allowed exception categories, or missing authoritative policy.
Escalation triggers	Policy conflict or absence; identity/order mismatch; customer disputes decision after explanation; special category not supported; suspected abuse; customer requests human.
Successful outcome	The customer receives a policy-grounded decision and understands the reason and next option.

Main customer flow

1. Identify the order and item.
2. Ask for the return reason and condition only when needed.
3. Retrieve the policy version applicable to the item, seller, location, and purchase date.
4. Evaluate return window, exclusions, condition, and special-category rules.
5. Show eligible or ineligible status with reason, deadline, and available next actions.
6. Offer to create a return when eligible or request human review when an exception may apply.

Alternative and failure flows

- Final-sale item: explain the exclusion and whether damaged-item support is still available.
- Outside normal window: deny standard return but offer human review only when an exception category exists.
- Policy conflict: do not choose silently; escalate with both sources identified.
- Missing policy: explain that eligibility cannot be confirmed and escalate rather than guessing.
- Marketplace or third-party seller: use the seller-specific policy and identify the responsible party.

Acceptance criteria

PASS The decision uses the policy effective for the order, not simply the newest policy.

PASS The explanation states the decisive rule, deadline, or exclusion.

PASS The system does not create a return during an eligibility-only request.

PASS A policy conflict is visible and routed for review.

PASS An eligible result offers a direct transition to the create-return journey.

6. Create a return

Journey purpose

Create a verified return authorization and give the customer everything needed to send or drop off the item.

Contract area	Fixed decision
Preconditions	Customer and order are verified; item is return-eligible; no existing active return conflicts with the request.
Required inputs	Selected item(s), quantity, reason, preferred resolution, return method, and confirmation.
Customer confirmation	Required immediately before creating the authorization. The preview must show items, quantity, resolution, method, fee, and deadline.
Human approval	No human approval for a standard eligible return. Approval is required only for configured policy exceptions or unusually high-risk cases.
Escalation triggers	Return-service failure after safe retry; conflict with existing return; policy exception; unsupported item; fraud or abuse signal; customer requests human.
Successful outcome	A unique return authorization exists and the customer receives a complete return receipt and instructions.

Main customer flow

1. Select eligible item or items and quantities.
2. Collect return reason and optional notes.
3. Show permitted resolution methods: original payment, store credit, or exchange when available.
4. Show return method, fees if any, deadline, and expected refund timing.
5. Display a final action preview.
6. Obtain explicit customer confirmation.
7. Create the return authorization.
8. Generate or retrieve the label or drop-off instructions.
9. Display return ID, deadline, instructions, and next expected state.

Alternative and failure flows

- Partial quantity return: show exactly which units are included.
- Label generation failure: keep the return state consistent, retry safely, and provide progress or escalation.
- Existing return: show its status rather than creating a duplicate.
- Return fee: display before confirmation and include in the preview.
- Return method unavailable: offer another permitted method or human review.

Acceptance criteria

- PASS** No return is created before explicit confirmation.
- PASS** A repeated confirmation or retry does not create a duplicate return.
- PASS** The receipt includes return ID, items, quantities, resolution method, deadline, and instructions.
- PASS** A label failure does not produce an incorrect success message.
- PASS** Existing active returns are detected and shown.

7. Damaged-item replacement

Journey purpose

Collect sufficient evidence, validate eligibility and inventory, and create a safe replacement or route the case for approval.

Contract area	Fixed decision
Preconditions	Customer and order are verified; selected item was fulfilled; replacement journey is supported for the product category.
Required inputs	Order and item, damage description, preferred resolution, image evidence when required, replacement address confirmation, and final confirmation.
Customer confirmation	Required before creating the replacement. The preview must show item, quantity, shipping address, return requirement, and expected timing.
Human approval	Automatic only for low-risk cases at or below \$250. Above \$250, policy exception, changed address after restricted stage, unclear evidence, or risk signal requires human approval.
Escalation triggers	Customer requests human; missing or contradictory policy; high value; risk signal; unavailable inventory with no supported alternative; evidence remains insufficient; tool failure; address concern.
Successful outcome	A replacement is reserved and confirmed, or a clearly described approval/review state exists without duplicate actions.

Main customer flow

1. Identify the order and damaged item.
2. Collect a concise damage description.
3. Request image evidence when item value is above \$75 or the policy requires it.
4. Check damaged-item policy, prior claims, risk, and replacement eligibility.
5. Check replacement inventory and permitted alternatives.
6. Confirm the destination address and whether a return of the damaged item is required.
7. Show replacement preview, including item, address, return obligation, and expected timing.
8. Obtain customer confirmation.
9. Create automatically when low-risk and value is \$250 or less; otherwise request human approval.
10. Provide a claim or replacement reference and status.

Alternative and failure flows

- No inventory: offer refund, waitlist/backorder where supported, equivalent replacement, or human assistance.
- Evidence unclear: ask once for a clearer image or additional detail, then escalate rather than repeatedly blocking the customer.
- Replacement address differs from order: require re-verification; after dispatch-related restrictions, require human approval.
- High fraud or repeated-claim signal: do not accuse the customer; move to review with neutral language.
- Return required: provide label and deadline as part of the replacement receipt.
- Partial failure after reservation: show pending/recovery state and do not create another replacement on retry.

Acceptance criteria

PASS Evidence requirements follow item value and policy.

PASS A replacement is not created before confirmation and required approval.

PASS Retries do not create duplicate replacement orders.

PASS Inventory is verified before success is claimed.

PASS The receipt states whether the damaged item must be returned and provides instructions when required.

PASS High-risk review language remains neutral and customer-friendly.

8. Refund request

Journey purpose

Determine the permitted refund, show the exact effect, and perform it only within the customer, policy, and approval boundaries.

Contract area	Fixed decision
Preconditions	Customer and order are verified; refundable transaction and permitted reason are identified; refund tool is available.
Required inputs	Order or transaction, reason, refund amount or eligible items, refund method, and confirmation.
Customer confirmation	Always required immediately before the refund is submitted or approval request is created.
Human approval	Automatic only at or below \$100 and low risk. \$100.01 to \$500 requires human approval. Above \$500 requires full human takeover. Any policy exception or risk signal requires review.
Escalation triggers	Customer asks for human; amount exceeds automation; policy exception; payment mismatch; duplicate or unknown status; fraud signal; service failure; customer disputes calculation.
Successful outcome	The refund system confirms a refund or a review state exists with a clear amount, destination, timing, and reference.

Main customer flow

1. Identify the transaction or returned item.
2. Check refund eligibility, previous refunds, payment method, and amount.

3. Calculate the permitted refund and disclose fees, deductions, or exclusions.
4. Show amount, destination, expected timing, and consequences.
5. Obtain explicit customer confirmation.
6. For low-risk refunds of \$100 or less, perform the refund automatically.
7. For refunds above \$100, create an approval request and show pending status.
8. After the service confirms the action, display reference, amount, method, status, and timing.

Alternative and failure flows

- Partial refund: show item-level or charge-level breakdown.
- Refund already issued: show existing status and reference rather than creating another.
- Payment method unavailable: explain the permitted alternative and require confirmation if it changes destination.
- Service timeout after submission: treat as unknown/pending, reconcile status, and never blindly retry a financial action.
- Amount disagreement: explain calculation and offer human review.

Acceptance criteria

PASS The amount in the confirmation preview matches the submitted amount.

PASS The product never reports completed when status is pending or unknown.

PASS Duplicate refunds are prevented using existing-status checks and safe request handling.

PASS The refund destination and expected timing are displayed.

PASS Threshold and risk rules correctly trigger approval or takeover.

9. Subscription cancellation

Journey purpose

Allow a customer to end a subscription transparently while preserving choice and avoiding manipulative retention behavior.

Contract area	Fixed decision
Preconditions	Customer is authenticated; active subscription and contractual cancellation options are available.
Required inputs	Subscription selection, cancellation timing, optional reason, retention decision, and confirmation.
Customer confirmation	Required before cancellation. The preview must show effective date, access end date, expected charges or credits, and any irreversible consequence.
Human approval	No human approval for a standard permitted cancellation. Human review is required for contract exceptions, special credits, account ownership uncertainty, or disputed balances.
Escalation triggers	Customer asks for human; contract exception; identity uncertainty; service failure; cancellation not supported; legal or billing dispute; policy conflict.
Successful outcome	The subscription system confirms the cancellation timing and the customer understands remaining access and billing.

Main customer flow

1. Show current plan, renewal date, price, and cancellation options.
2. Explain immediate versus end-of-term consequences when both are available.
3. Ask for a reason only when useful and optional unless policy requires it.
4. Present at most one permitted retention option when applicable.
5. Allow the customer to decline without repeated persuasion.
6. Show final cancellation preview and access/billing consequences.
7. Obtain confirmation.
8. Cancel through the subscription service.
9. Display effective date, remaining access, final billing information, and reference.

Alternative and failure flows

- Annual commitment or contract: explain contractual restriction and route allowed exceptions to human review.
- Outstanding balance: explain whether cancellation is allowed and what remains due.
- Already scheduled cancellation: show current state and allow permitted reversal or update.
- Service failure: preserve the customer decision, show pending/recovery state, and escalate when needed.
- Retention offer not permitted: skip it entirely.

Acceptance criteria

PASS The product offers no more than one retention option.

PASS Declining the offer proceeds directly to cancellation without additional friction.

PASS The cancellation preview is explicit about effective date and billing.

PASS Success is displayed only after subscription-service confirmation.

PASS Existing scheduled cancellations are detected and not duplicated.

10. Human escalation

Journey purpose

Move a conversation to the correct human queue with complete, useful context and a clear customer expectation.

Contract area	Fixed decision
Preconditions	A customer request, risk rule, unsupported condition, approval requirement, or repeated failure indicates that automation should stop or pause.
Required inputs	Conversation, customer identity when available, journey, urgency, risk, summary, evidence, tool history, workflow state, and reason.
Customer confirmation	No customer confirmation is required to honor an explicit human request. The agent may confirm before escalating only when clarifying the requested team or issue will reduce misrouting.
Human approval	The human receiving the case may need authority for the requested action. The routing decision itself does not require separate approval.
Escalation triggers	Explicit human request; low confidence; high-risk action; approval threshold; policy conflict; unsupported request; identity failure;

Contract area	Fixed decision
	repeated tool failure; legal, security, fraud, or abuse concern; repeated customer disagreement.
Successful outcome	A support case or approval request is assigned or queued with complete context, customer-visible status, and no need to restart the story.

Main customer flow

1. Recognize a customer request for a person or an internal escalation trigger.
2. Stop risky or repeatedly failing automated actions.
3. Create a concise handoff summary and identify pending decisions.
4. Select the correct queue and priority.
5. Create or update the support case.
6. Tell the customer why the handoff is happening in appropriate language.
7. Show expected next step or waiting state.
8. Allow a representative to join and take control.

Alternative and failure flows

- No representative immediately available: show queued state and preserve conversation continuity.
- Case creation fails: retry safely, maintain visible pending state, and provide alternative contact guidance when configured.
- Customer is anonymous: transfer general context while clearly marking identity as unverified.
- Approval-only escalation: keep the agent conversation active while the representative approves or rejects the pending action.
- Customer disconnects: preserve case and allow later resume.

Acceptance criteria

PASS The handoff contains transcript, summary, customer request, journey, policies, tool history, failures, workflow state, and recommended next action.

PASS The customer is told whether a representative is joining now or the case is queued.

PASS An explicit request for a human is honored without repeated deflection.

PASS The case is routed to a queue matching the journey and risk.

PASS The representative can continue from the existing conversation context.

11. Approval boundaries and risk rules

Policy model

Thresholds are configurable per tenant, but V1 uses fixed demo defaults so the product can be designed and tested consistently.

Action	AI autonomy	Customer confirmation	Human approval or takeover
View order or shipment	Allowed after appropriate identity check.	No	No

Action	AI autonomy	Customer confirmation	Human approval or takeover
Explain policy or eligibility	Allowed when grounded in approved policy.	No	Only for conflict, missing policy, or exception request
Create standard return	Allowed when eligible and low risk.	Required	Only for exception or risk
Refund \$100 or less	Allowed when low risk and verified.	Required	No
Refund \$100.01-\$500	May prepare and submit for approval.	Required	Required
Refund above \$500	May collect context but must stop before action.	Required before handoff action	Full human takeover
Replacement \$250 or less	Allowed when low risk, evidence complete, and inventory available.	Required	No
Replacement above \$250	May prepare request.	Required	Required
Change address before fulfillment cutoff	Allowed after re-verification and policy check.	Required	No unless risk
Change address after restricted fulfillment stage	Not autonomous.	Required	Required
Cancel standard subscription	Allowed when contract permits.	Required	No
Contract exception, special credit, fraud, legal, or security case	Not autonomous.	As appropriate	Required

11.1 Universal risk triggers

- Identity mismatch or insufficient authentication.
- Repeated claims, duplicate requests, inconsistent evidence, or other configured abuse signals.
- Policy conflict, missing authoritative policy, or requested exception.
- Unknown result after a consequential action was submitted.
- Tool or business-system failure after safe retry.
- Customer states legal, security, fraud, discrimination, self-harm, medical, or other sensitive concern requiring a specialized human process.
- Customer explicitly requests a human.

12. Product screen inventory and screen contracts

12.1 Customer application

Screen	Required content and actions
Chat home / launcher	Brand identity, supported-purpose summary, start conversation, privacy notice, and access to prior conversations when signed in.
Conversation screen	Messages, typing/streaming state, source references, suggested actions, attachment button, human request, and clear error/progress states.
Identity verification	Reason verification is needed, permitted verification methods, success/failure, retry limits, and cancel/back behavior.
Order and item selector	Recent orders, dates, item names, status, thumbnails when available, search/filter, and selected state.

Screen	Required content and actions
Evidence upload	Requested evidence, supported formats, preview, remove/replace, upload progress, privacy explanation, and failure recovery.
Action review and confirmation	Action type, target item/account, amount or effect, fees, destination, timing, policy note, confirm, cancel, and edit/back.
Workflow progress	Current state, completed steps, waiting reason, expected next event, notifications, and human-support option.
Escalation state	Reason in customer-friendly language, queue or representative status, expected next step, and uninterrupted transcript.
Conversation history	Past conversations, status, date, journey, resolution summary, and resume when permitted.

12.2 Human support workspace

Screen	Required content and actions
Escalation inbox	Queue, priority, waiting time, customer, journey, risk, sentiment, assignment, and filters.
Case detail	Transcript, AI summary, customer/account context, journey state, evidence, internal notes, and customer-visible messages.
Activity timeline	Knowledge consulted, policies checked, tool calls, results, failures, confirmations, approvals, and state changes.
Approval panel	Requested action, amount/effect, policy, risk reasons, evidence, approve, reject, modify, and required reason.
Human conversation panel	Join, send, use/edit suggested response, attach content, and return control to AI.
Resolution panel	Outcome, action reference, customer message, close reason, feedback classification, and create-regression option.

12.3 Minimum operations and administration screens

Screen	Required content and actions
Agent overview	Agent status, supported journeys, environment, current release, health, usage, and edit access.
Journey configuration	Purpose, required information, policies, permitted actions, confirmation, approval, completion, escalation, and examples.
Knowledge sources	Sources, status, sync time, version, effective dates, conflicts, failures, and retrieval preview.
Policy and thresholds	Return rules, refund/replacement limits, evidence requirements, approvals, authentication, and exception routing.
Human queues	Queue purpose, membership, hours/status, accepted journeys, priorities, and approval authority.
Users and roles	User, role, environment access, queue membership, and status.

13. Cross-product states, notifications, and experience principles

13.1 Fixed customer-visible states

State	Meaning
Collecting information	The product needs specific information to progress.
Checking	The product is retrieving or validating data.
Ready for confirmation	The exact proposed action is displayed and no action has occurred yet.
Processing	A confirmed action is being submitted.
Completed	The business system confirmed completion and a reference is available.
Pending external event	The workflow is waiting for customer, human, carrier, warehouse, inventory, or another system.
Needs human review	Automation has paused and a human decision or action is required.
Failed - recoverable	A temporary or retryable failure occurred; the product explains recovery.
Unable to complete	The journey cannot continue automatically and provides escalation or alternative guidance.

13.2 Notification contract

- V1 uses in-product conversation updates. Email, SMS, or push notifications are post-v1.
- A pending workflow shows what it is waiting for and whether the customer can leave safely.
- When a human joins, the customer sees the change in control.
- When an action completes later, the conversation receives an auditable completion update.

13.3 Experience principles

- Resolution over conversation: every journey should move toward a business outcome.
- Truthful status: never imply an action succeeded before verification.
- Customer control: preview, confirm, cancel, and request a human.
- Minimal repetition: preserve context across steps and handoff.
- Explain important decisions: show the governing reason without overwhelming the customer.
- One clear next step: avoid presenting many irrelevant options.
- Neutral risk language: do not accuse customers when a case requires review.
- Accessible and responsive interfaces: keyboard navigation, clear labels, readable status, and mobile-compatible layouts.

14. V1 acceptance summary and readiness for technical design

14.1 Product-level acceptance criteria

PASS All seven journeys have a normal path, alternative path, failure path, and escalation path.

PASS Every consequential action has a preview, explicit confirmation, verified result, and audit-visible reference.

PASS Refund and replacement thresholds trigger the correct autonomous, approval, or takeover behavior.

PASS The customer can request a human at any point and does not need to repeat the issue after handoff.

PASS The human workspace shows the complete context required to continue the case.

PASS The product distinguishes completed, pending, review-required, recoverable failure, and unable-to-complete states.

PASS Policy-based decisions identify the governing rule or source.

PASS Repeated requests do not create duplicate returns, refunds, replacements, cancellations, or cases.

PASS Operations users can configure the V1 thresholds, knowledge, journeys, queues, and permissions represented in this contract.

PASS The product scope remains web-chat and e-commerce focused until these criteria are met.

14.2 Decisions now fixed for technical design

Decision area	Fixed input to technical design
Domain and channel	E-commerce support through web chat.
Primary experiences	Customer interface, human workspace, and minimum operations/admin controls.
Journeys	Tracking, eligibility, return creation, damaged replacement, refund, cancellation, and handoff.
Autonomy	Low-risk actions within defined thresholds; confirmation for consequential actions; approval or takeover above boundaries.
Human model	Queue-based handoff and approval with complete shared context and resumable conversation.
Outcome truth	Business-system confirmation is required before completed status.
Product states	Collecting, checking, confirmation, processing, completed, pending, review, recoverable failure, unable to complete.
Scope boundary	No voice, additional domains, free-form builder, or autonomous high-risk financial actions in V1.

Ready for the next stage

Technical design can now begin from a stable product contract: system boundaries, architecture, data ownership, agent orchestration, retrieval, tool contracts, workflow durability, security, observability, scaling assumptions, and implementation phases should all trace back to the decisions in this document.